

Can we turn 7 years of public IRS data into decisions Fairlight can act on tomorrow?

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The Problem



Which organizations are worth backing?

Funders guess where to allocate philanthropic capital without data.



Can we make equal comparisons?

Nonprofit leaders can't benchmark themselves against true peers.



Can it advise clients on organizations' financial health while avoiding the time cost of pursuing spreadsheets?

Time is critical, both for historical understanding and present decisions

400K+ Form 990s filed across 7 years. The data is public. And yet, decisions to invest remain guesswork.

Nonprofit Sector Is More Fragile Than It Looks

25.4%

At-risk rate across 130,000 unique organizations

1 in 4 organizations are categorized in financial distress right now. But why?

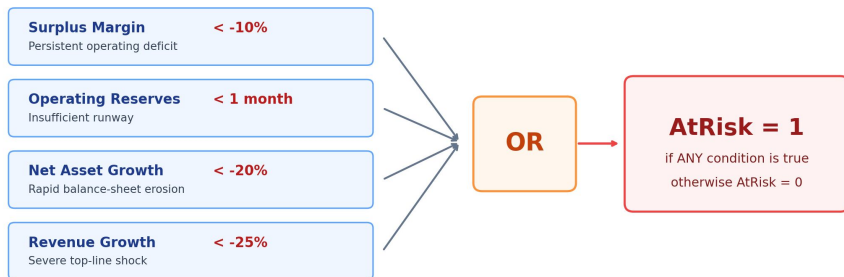
Two Major Contributing Reasons:

1. Debt Ratio
 - Funding fluctuates, debt doesn't
2. Contribution Growth/Shrinkage
 - Shrinking contributions = early warning donors and grants are pulling back

At-Risk Score

At-Risk Label Logic

Binary alert: flag an org when any critical stress signal appears



Why this works

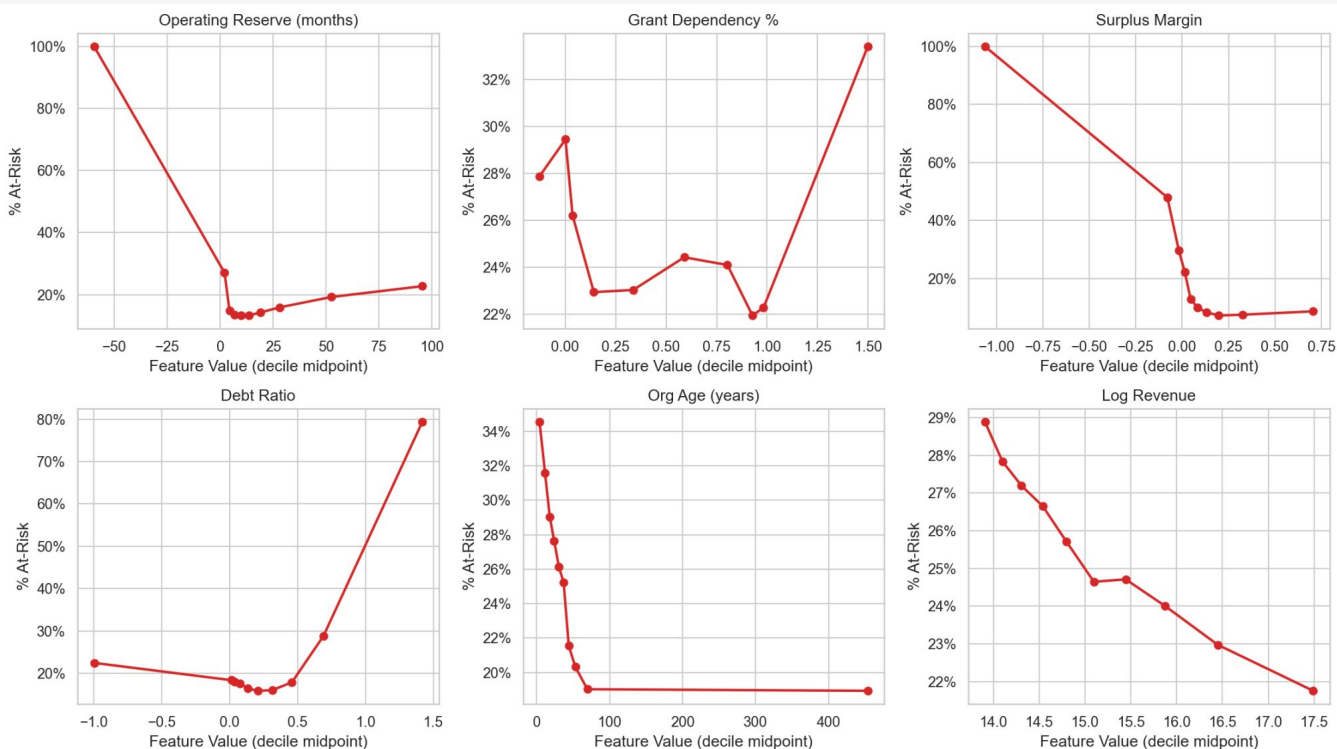
- Covers 4 chief stress indicators
- Transparent and easy to explain
- Useful as an early warning alert

Component Definitions

- **SurplusMargin:** Measures whether the nonprofit is consistently earning more than it spends, with negative values signaling operating deficits.
- **OperatingReserveMonths:** Measures cash runway as months of expenses covered by net assets, with low values meaning weak short-term resilience.
- **NetAssetGrowthPct:** Measures year-over-year change in net assets, where large declines indicate structural financial deterioration.
- **RevenueGrowthPct:** Measures year-over-year revenue momentum, where sharp drops indicate funding instability and elevated risk.

Model selection and why is it Random Forest

Each plot shows a financial metric's relationship to risk, and they are not straight. Risk spikes and drops at critical thresholds



A linear Debt Ratio graph would argue both debt and risk increase proportionally, but logically and visually the relationship is curved; exponential. Trees split on thresholds, they are built to capture irregular patterns.

Tested 4 models, and with a high ROC AUC for Random Forest, the EDA presented the path to the model

The Resilience Score

Think of it as credit scores for nonprofit financial health.

Component	Weight	Why — grounded in EDA findings
Operating Reserves	30 %	Top stability signal
Revenue Diversification	20 %	Counters grant dependency
Program Efficiency	20 %	Core mission metric
Surplus Margin	15 %	Deficit is an at-risk trigger
Low Debt	15 %	Orgs carrying a large debt are more susceptible to funding shocks

Strong ≥ 70

Moderate 40–69

At Risk < 40

**We turned these findings
into six decision-making
tools.**

01 Executive Overview

02 Peer Benchmarking

03 Resilience Explorer

04 Stress Test Simulator

05 Hidden Gems Finder

06 Brand Map

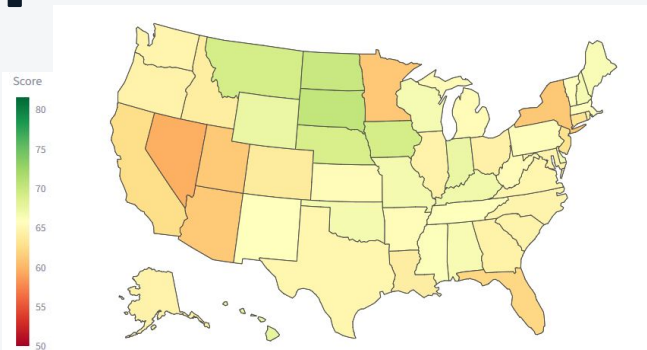
Executive Overview - Nonprofit Financial Resilience Dashboard

Why we built it

Funders need a sector-wide scan before zooming in. EDA showed risk is unevenly distributed — you need the full picture first.

What it shows

KPI cards (at-risk rate, median reserves), resilience distribution, state choropleth, sector ranking bar chart.



Nonprofit Financial Resilience Dashboard

A data-driven look at the financial health of 396,910 U.S. nonprofits across 59 states and territories, using IRS Form 990 filings from 2018 – 2024.

Total Nonprofits Analyzed

396,910

At-Risk Rate

25.4%

Financially Strong (score 70+)

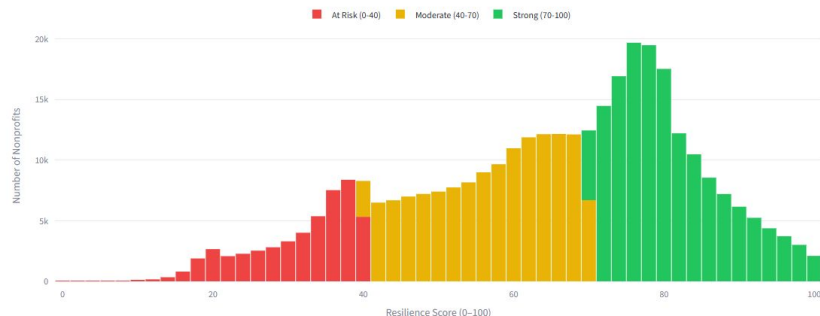
39.7%

Median Months of Reserves

11.4

Key finding: About 1 in 4 nonprofits show signs of financial distress — running large deficits, shrinking revenue, or dangerously low reserves. Targeted support could prevent closures.

How Resilient Are U.S. Nonprofits?



What is the Resilience Score?

A 0–100 composite rating based on five financial fundamentals:

- Operating reserves (30 pts)
- Revenue diversification (20 pts)
- Program spending efficiency (20 pts)
- Surplus margin (15 pts)
- Low debt (15 pts)

Higher scores mean the organization is better positioned to weather funding disruptions.

Peer Benchmarking

Why we built it

Raw metrics mean nothing without something to compare against; similar organizations or sector averages.

What it shows

How an org compares against its sector's medians across 5 key metrics.

Organization Snapshot

#2ND CHANCE PROJECT

Location: WOODBURY, MN | Sector: Housing & Shelter | Size: 1M-5M | Peer group: 330 similar organizations

Search for an organization

#2ND CHANCE PROJECT

61/100

Moderate

How Does This Organization Compare?



Metric-by-Metric Breakdown

- Program Spending %: 1.00 vs. peer median 0.83 (top 1%)
- Fundraising Cost %: 0.00 vs. peer median 0.00 (top 73%)
- Surplus / Deficit Margin: 0.26 vs. peer median 0.09 (top 20%)
- Months of Reserves: 4.25 vs. peer median 11.93 (top 71%)
- Debt-to-Asset Ratio: 0.00 vs. peer median 0.47 (top 99%)

● = better than peers | ● = below peers

Resilience Explorer

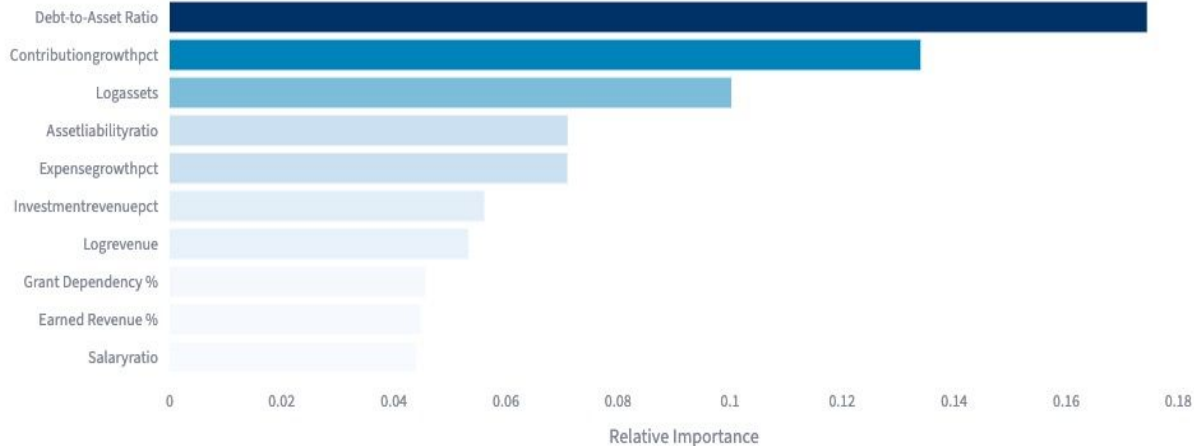
Why we built it

Nonprofit funders need to understand which performance indicators are the strongest signals of financial resilience.

What it shows

Top performance indicators in order of relative importance.

What Drives Financial Resilience?



Key takeaway:

The three strongest predictors of financial resilience are **Debt-to-Asset Ratio**, **Contributiongrowthpct**, and **Logassets**.

Organizations that maintain healthy surpluses, steady revenue growth, and adequate cash reserves are far more likely to survive funding disruptions.

Resilience Explorer

Why we built it

While sector wide analyses are useful, it is also important to understand metrics at an organizational level.

What it shows

Risk probability as calculated by our model of an org compared to similar orgs in the same sector, revenue size, or state.

Filter by sector

All Sectors



Filter by Revenue

All Sizes



Filter by state

All States



Browse organizations:

	Organization	State	Sector	Size	Resilience Score	At Risk?	↑ Risk Probability
383106	CLARENCE E MCLENDON MINISTRIES FULL HARVEST INTERNATIONAL CHURCH	CA	Religion	1M-5M		13 No	17.5%
29696	INLAND PORT TRANSPORTATION MANAGEMENT ASSOCIATION	TX	Community Improvement	1M-5M		12 No	19.2%
395028	Greater Wichita Partnership Inc	KS	Community Improvement	1M-5M		12 No	20.8%
210359	Florida High School For Accelerated	FL	Education	5M-10M		12 No	21.7%
251050	DENVER VOLUNTEER FIRE DEPARTMENT INC	NC	Healthcare	1M-5M		9 No	24.2%
209389	Harding Fine Arts Academy Inc	OK	Education	1M-5M		12 Yes	31.7%
252890	FIREFIGHTERS CHARITABLE FOUNDATION INC	NY	Healthcare	1M-5M		9 Yes	44.2%
173733	VOLUNTEER FIREFIGHTER ALLIANCE INC	TN	Public Safety & Disaster Relief	5M-10M		14 Yes	49.2%
127652	Harding Fine Arts Academy Inc	OK	Education	1M-5M		5 Yes	58.3%
184834	Hope Lives Vive La Esperanza	AZ	Human Services	1M-5M		5 Yes	70.8%

Stress Test Simulator

Why we built it

Knowing an org is 'At-Risk' isn't enough — funders need to know the effects of a specific shock.

What it shows

Specific shock type and magnitude and the effect size among different sectors of the nonprofit industry.

Customize Shock Intensity

✓ Sync all sliders — drag one to update all streams equally

Donations & grants drop

20%

Government grants drop

20%

Program revenue drop

20%

Investment income drop

20%

Overall Impact

Critical

36,430

↑ 9.2%

At Risk

36,057

↑ 9.1%

Stressed

233,574

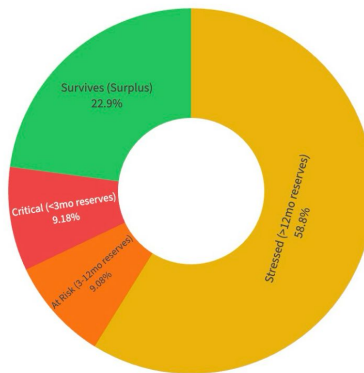
↑ 58.8%

Survives

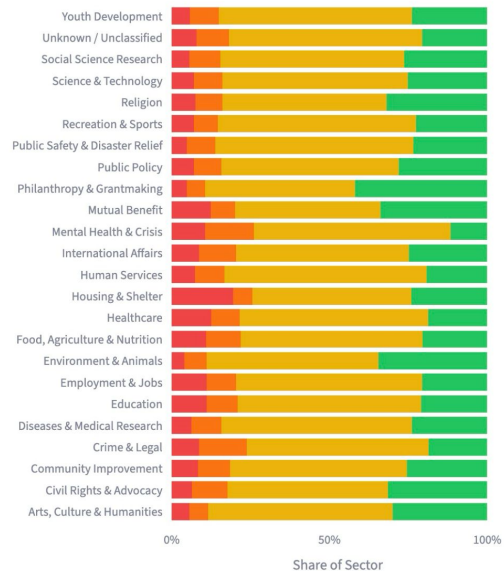
90,849

↑ 22.9%

Impact: Under this scenario, 18.3% of nonprofits would face serious financial distress (critical or at risk). That's roughly 72,487 organizations that could struggle to maintain services. On average, nonprofits would lose 22.6% of their total revenue (100% of organizations are directly exposed to this shock).



■ Stressed (>12mo reserves) ■ Survives (Surplus)
■ Critical (<3mo reserves) ■ At Risk (3-12mo reserves)



Stress Test Simulator

Why we built it

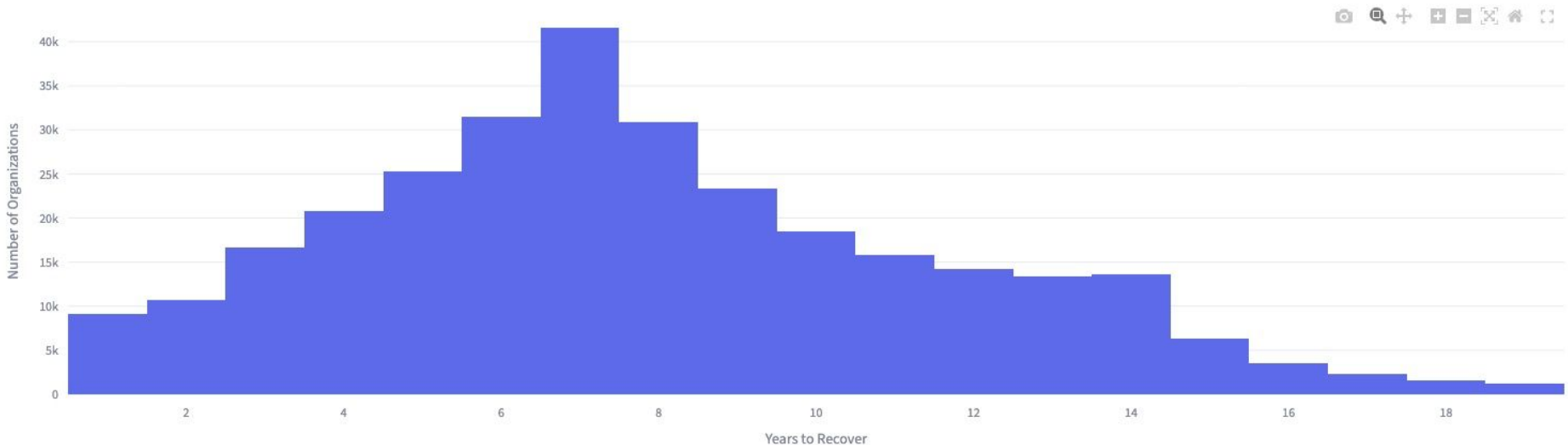
Funders can see which sectors are most prone to certain funding shocks and how affected they will be in terms of recovery time.

What it shows

The distribution of recovery times for nonprofits given a certain type of funding shock and magnitude. In this case, for a 20% shock to all funding sources, the majority of orgs will take 7-8 years to recover.

Recovery Timeline

For organizations that would go into deficit, the **median recovery time is 7.0 years** (average: 7.9 years), assuming a 5% annual revenue recovery rate.



Stress Test Simulator

Why we built it

Aggregate risk identifies the problem; the org view makes it actionable.

What it shows

Per-org shock diagnostics: pre/post revenue by stream, tier downgrade, time to insolvency, and recovery horizon.

Select an Organization

Filter by sector

All sectors

Organization

DISABILITY RIGHTS SOUTH

DISABILITY RIGHTS SOUTH

Civil Rights & Advocacy · SC · 1M-5M

Current **Survives** → Post-shock **Stressed** + Downgraded 1 tier

Revenue Lost

\$1.1M

↓ -35.7%

Post-Shock Revenue

\$1.9M

↓ -35.7% vs pre-shock

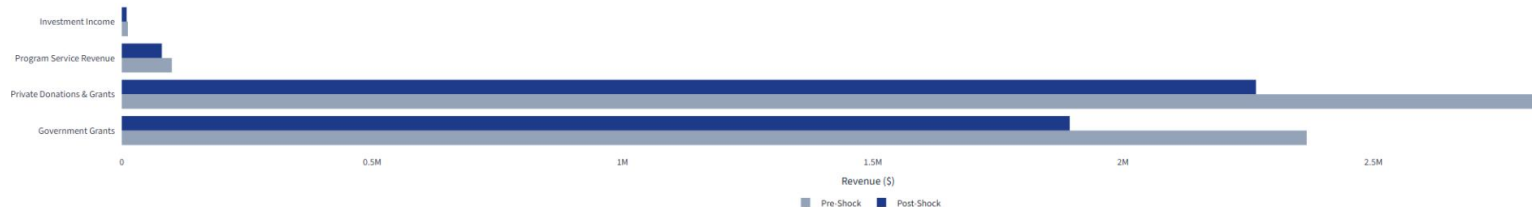
Months to Insolvency ⓘ

12.9 mo

Recovery Time ⓘ

11.0 yrs

Revenue Stream Impact



Why this org is affected: The shock hits **Private Donations & Grants** (19.0% of total revenue, \$566K lost) and **Government Grants** (15.9% of total revenue, \$473K lost) and **Program Service Revenue** (0.7% of total revenue, \$20K lost) and **Investment Income** (0.1% of total revenue, \$2K lost). That's a 35.7% total revenue hit, which drives the status change above.

Hidden Gems Finder

Why we built it

Risk is only half the story. This view surfaces nonprofits that are both high-impact and financially credible.

What it shows

Ranked, decision-ready shortlists.
DonationToStabilize metric — exact dollars to reach 6-month reserve. Top gem cards as well as a detailed list.

Choose a Prioritization Tier

☒ Top 25 — Immediate funding focus · 25 orgs ☐ Top 100 — Priority shortlist · 100 orgs ☐ Top 500 — Extended shortlist · 500 orgs ☐ Full universe (all gems) · 3,620 orgs

Organizations in Tier

25

Total Funding to Stabilize ⓘ

\$2.6M

Annual Mission Impact Protected ⓘ

\$52.6M

Portfolio ROI ⓘ

20.0 / 1

How this shortlist is built: every candidate is ranked on *Cost-Efficiency Score* (0-100) — a weighted blend of *mission ROI* (40%), overall impact quality (30%), revenue growth (15%), and funding urgency (15%). Median mission ROI in this tier: \$22.8 of annual program activity protected per \$1 donated.

Refine Your Shortlist

State

All States

Sector

All Sectors

Max donation per org

ⓘ
\$430000

Showing 25 organizations matching your filters

Taylor County Emergency Squad

GRAFTON WV · Healthcare

#1

MISSION ROI

\$33.6 of programs / \$1 donated

Cost-Eff. Score

88/100

🔥 \$399K to reach 6-month reserves

Resilience

Moderate

📊 Score Breakdown

OVERFLOW HEALTH ALLIANCE INC

Jacksonville FL · Education

#2

MISSION ROI

\$12.9 of programs / \$1 donated

Cost-Eff. Score

88/100

🔥 \$428K to reach 6-month reserves

Resilience

Strong

📊 Score Breakdown

HAWAIIAN CANOE RACING ASSOCIATION

KIHEI HI · Education

#3

MISSION ROI

\$25.3 of programs / \$1 donated

Cost-Eff. Score

82/100

🔥 \$36K to reach 6-month reserves

Resilience

Strong

📊 Score Breakdown

Hidden Gems Finder

Why we built it

Risk is only half the story. This view surfaces nonprofits that are both high-impact and financially credible.

What it shows

Allows for a deeper breakdown at a company level to understand the specific metrics that help classify it as a “hidden gem”

Ranked List

	Rank	Organization	State	Sector	Cost-Eff. Score	ROI (\$ programs / \$1)	Donation to Stabilize	Annual Program Impact	Impact Score	Resilience	Reserve Months	Revenue Growth
0	1.000000	Taylor County Emergency Squad	WV	Healthcare	88	\$33.55	\$399,208	\$13,394,476	77	61	2.9	+101.9%
1	2.000000	OVERFLOW HEALTH ALLIANCE INC	FL	Education	88	\$12.88	\$428,446	\$5,519,025	85	74	2.2	+102.6%
2	3.000000	HAWAIIAN CANOE RACING ASSOCIATION	HI	Education	82	\$25.30	\$35,906	\$908,486	82	74	5.5	+346.3%
3	4.000000	EEL RIVER RECOVERY PROJECT INC	CA	Environment & Animals	82	\$14.65	\$106,749	\$1,563,561	88	83	5.2	+500.0%
4	5.000000	BHS INC	MT	Recreation & Sports	82	\$24.24	\$55,793	\$1,352,688	77	78	5.5	+500.0%
5	6.000000	HAPPY CANYON COMPANY INC	OR	Human Services	82	\$23.96	\$65,603	\$1,572,050	77	65	5.6	+500.0%
6	7.000000	Najam Institute Foundation Inc	GA	Human Services	81	\$36.75	\$47,023	\$1,728,298	80	80	5.7	+107.9%
7	8.000000	USACANADA LIONS LEADERSHIP FORUM	WV	Education	81	\$22.82	\$34,367	\$784,404	87	82	5.5	+105.0%
8	9.000000	CENTER FOR EMOTIONAL WELLNESS PC	TX	Mental Health & Crisis	81	\$26.60	\$28,073	\$746,801	75	74	5.6	+500.0%
9	10.000000	PENINSULA BALLET THEATRE	CA	Arts, Culture & Humanities	81	\$18.24	\$59,461	\$1,084,634	77	65	5.3	+412.1%
10	11.000000	WEST-MONT UNITED SOCCER ASSOCIATION	PA	Human Services	81	\$29.07	\$75,175	\$2,185,117	81	66	5.6	+89.2%

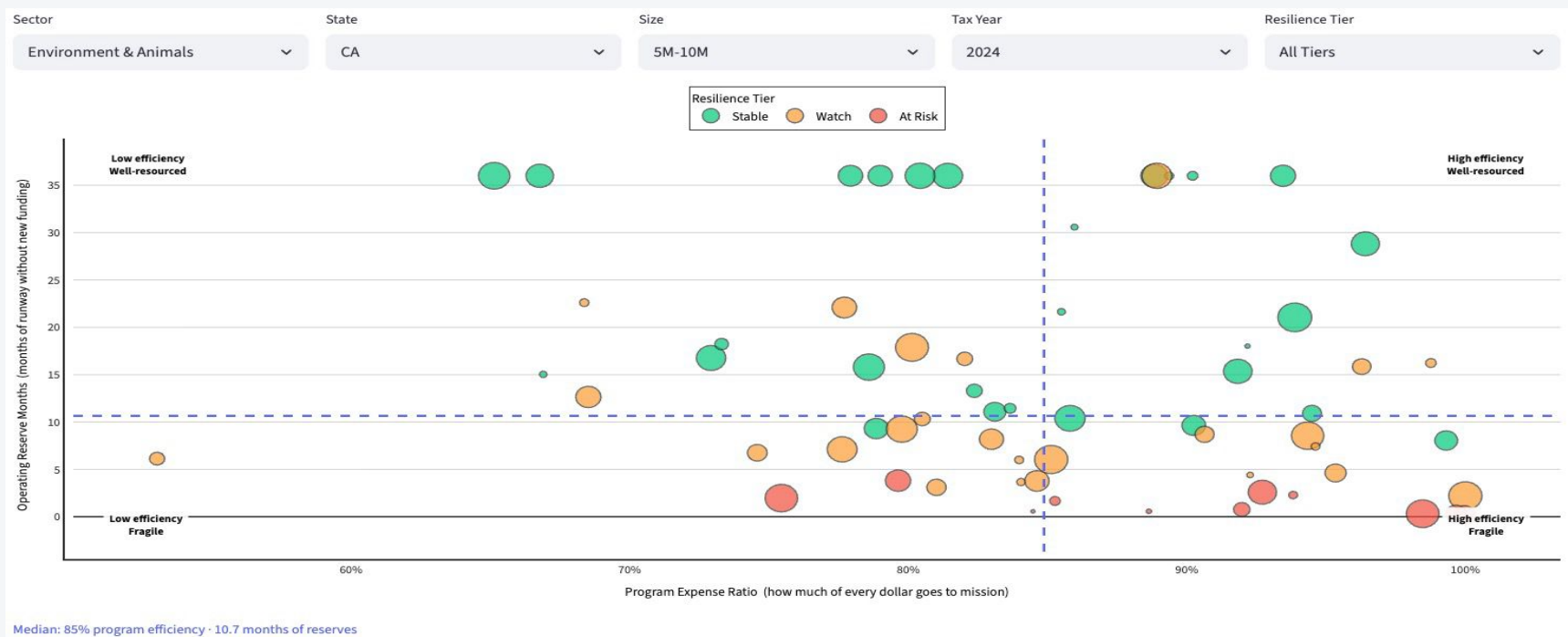
Brand Map

Why we built it

Gives users a single view of the entire sector across two important dimensions: Efficient org mission delivery and operating reserve months.

What it shows

Visually compare all organizations within a sector in a single view.



Overview Summary

- 1 in 4 organizations are at risk across 130K unique organizations
- Our initial analysis led to understanding relationships between risk and metric

01 Executive Overview → big picture countrywide

02 Peer Benchmarking → apples to apples org comparison

03 Resilience Explorer → Financial resilience predictors

- Estimated continuous risk probability using ML
- Developed a quantified risk metric through Resilience Scoring

04 Stress Test Simulator → sector and org shock analysis

05 Hidden Gems Finder → ROI at individual level

06 Brand Map → Visual comparison of organizations